COMPARATIVE METABOLIC EFFECTS OF THREE TYPES OF COMBINED ORAL CONTRACEPTIVE PILLS IN CHINESE WOMEN

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Abstract

The results of a prospective longitudinal controlled study comparing some metabolic effects of a low-dose levonorgestrel-ethinyl oestradiol combined oral contraceptive (Microgynon® 30), a desogestrel-ethinyl oestradiol-containing pill (Marvelon®) and a levonorgestrel-ethinyl oestradiol triphasic preparation (Triquilar®) after one year of treatment in ethnic Chinese women are presented. Serum cholesterol levels are similar for the three pills at 6 and 12 months of treatment. Marvelon® and Triquilar® users showed rises in triglyceride levels while Microgynon® 30 users had no change after one year of treatment. Lipoprotein fractions showed favourable changes in Triquilar® and Marvelon® users with significant rise in HDL fractions at 6 and 12 months; Microgynon® 30 users had equivocal changes in HDL fractions. Carbohydrate metabolism as reflected by glucose tolerance showed deterioration with all 3 pills, being least with Microgynon® 30 and greatest with Marvelon® at 12 months

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Introduction

Since the late 1960s, the oestrogen component of the combined oral contraceptive (OC) has been linked to thromboembolism (1,2). It was only realised much later that the progestogen component too had a role in causing adverse cardiovascular effects with the publication of the Royal College of General Practitioners' study in 1977 (3). This interest gained momentum with Bradley et al.'s (4) work on the influence of progestogens on serum high density lipoprotein cholesterol and heart disease. The aetiology of cardiovascular disease is multifactorial, among which smoking, age, obesity, family history, diet, diabetes and genes play important roles. Synthetic oestrogens and progestogens can aggravate an already existing predisposition to heart disease by altering carbohydrate and lipoprotein metabolism. Both oestrogens and gonane progestogens (to which group belong all new progestogens) impair glucose tolerance by elevating blood sugar levels and increasing peripheral resistance to insulin action (5). Plasma triglycerides and phospholipids are elevated by oestrogen (5) while lipid and lipoprotein metabolism is affected by both oestrogens and progestogens often in a dose-dependent manner. These metabolic effects of oral contraceptives can potentially cause thrombosis and atherosclerosis. This realisation has lowered hormone content of the OC pill to microgram quantities (6,7). Further lowering of the progestogen dosage per cycle of use has been achieved with the 'phasic' approach resulting in the newer triphasic pills. The latest development is the discovery of newer progestogens like desogestrel, norgestimate and gestodene and initial indications (8) are that they have a better metabolic 'profile' than conventional OC pills. These studies have been in Caucasian women. Since race and diet can have profound influences on metabolic parameters, this study comparing the metabolic effects of a triphasic OC [Triquilar® = 30µg ethinyl oestradiol (EE) + 50µg levonorgestrel ((LNG) x 6 days; 40µg EE + 75µg LNG x 5 days; 30µg EE + 125µg LNG x 10 days] and an OC containing desogestrel (Marvelon® = 30µg EE + 150µg desogestrel) with a standard low-dose OC (Microgynon® = 30µg EE + 150µg LNG) on an ethnic Chinese population was conducted.

Patients and Methods

One-hundred-and-five Chinese women of proven fertility between the ages of 18 and 34 years seeking oral contraception at the University Department of Obstetrics and Gynaecology Fertility Control outpatient clinic were recruited for the study. All women admitted for study had regular menstrual cycles (cycle length 28 ± 4 days) and had not been on any form of hormonal contraception for the preceding six months. After obtaining informed consent, they were randomly allocated to one of three pill types, ie., Microgynon®30, Triquilar® or Marvelon®. They were instructed to take the pills as per manufacturer's recommendations and were followed-up at 3 months, 6 months and thereafter yearly. The subjects were encouraged to come earlier if necessary should they have any problems. The study period was one year but if the subjects found it suitable at the end of the study to continue with their pills, they were encouraged to do so.

Each volunteer acted as her own control. Blood sampling (fasting samples) was performed at pretreatment (control value) and at 6 months and 12 months of treatment. The samples were taken between days 22-24 of the menstrual cycle at the same time of the day, between nine and eleven in the morning after a half hour rest period prior to venepuncture.

Serum lipids, lipoproteins and glucose tolerance test were done at each blood sampling visit. For the glucose tolerance, blood glucose levels were measured at time 0 (fasting) and thereafter at 30 mins, 60 mins, 90 mins and 120 mins after a 50 gm oral glucose load.

Laboratory Methodology

Glucose Estimation

Blood samples were assayed for glucose (9) (glucose hexokinase method) using the Technicon Multianalyser.

Serum Lipoproteins

Preparative ultracentrifugation of serum was performed at 4°C and 80,000g for 22 hours using the gradient method according to Walton et al. (10). Three fractions were isolated, viz., very low density lipoproteins (VLDL), low density lipoproteins (LDL) and high density lipoproteins (HDL). Total and free cholesterol and triglycerides were estimated in each of the fractions with the methods described under serum lipids.

Serum Lipids

Total and free cholesterol were analysed according to the enzymatic method of Trinder (11) using Boehringer-Mannheim monotest and test combination kits, respectively. Triglycerides were estimated according to the method of Bucolo and David (12) using Boehringer-Mannheim test combination kit.

Quality Control

Quality control (internal quality control) in lipid assays were monitored by running 2 serum control pools (commercially available) (normal and high levels) in every assay. For total cholesterol, CV between assays at both normal (about 3 mmol/l) and high (about 8 mmol/l) was less than 2%. For free cholesterol, CV between assays at the low (less than 1 mmol/l) and high (above 2.5 mmol/l) ranges is 4 and 3%, respectively, while for triglycerides, at the normal level (about 2 mmol/l) CV is about 3% and at the higher limit (about 8 mmol/l) is about 5.5%. External quality control was assessed by an international quality control scheme (13) (International External Quality Assessment Scheme or IEQAS) conducted by WHO Collaborating Centre for Research and Reference Services in Clinical Chemistry, Birmingham, UK.

Statistical Method

Statistical analysis was carried out using paired t-test. The mean of pretreatment levels was designated as 100% and compared to post treatment levels after 6 and 12 months of oral contraception.

Results

Of the 105 subjects recruited, 34 were on Microgynon® 30, 39 on Triquilar® and 32 on Marvelon®. Following the strict criteria on inclusion and random allocation, all 3 groups were comparable on entry and showed no statistically significant differences in age, height and weight. All recruits were non-smokers. At the end of one year, there were still 68 women (Microgynon® 30 = 25; Triquilar® = 23; Marvelon® = 20) participating in the trial, the rest having dropped out from personal reasons or desire for pregnancy.

Changes in serum lipids at 6 and 12 months are shown in Table I. No appreciable change in serum triglycerides occurred with Microgynon®30, whereas statistically significant rises were seen with Triquilar® and Marvelon® to levels roughly 30-40% above pretreatment values. Serum cholesterol levels were no different from pretreatment levels for all 3 OC pills.

Triglyceride lipoprotein fractions (Table II) showed some interesting results. The standard OC Microgynon®30 does not seem to alter the various fractions over one year of treatment. Both Triquilar® and Marvelon® show rises in the various triglyceride lipoprotein fractions, with maximal rise in the HDL fraction.

Cholesterol lipoprotein fractions (free and total) which play an important role in the aetiology of heart disease showed (Table III) a cardioprotective potential for the newer OCs. Although the small numbers in this study make it difficult to make firm conclusions, the trend which has emerged makes for interesting comparison. Microgynon®30 causes a fall in both LDL and HDL (free and total cholesterol) liproproteins. The newer OCs Marvelon® and Triquilar® do not seem to affect VLDL levels whilst causing a roughly 10% fall in LDL (free and total cholesterol) levels at the end of one year of treatment with a corresponding rise of about 10% (Triquilar®) and 20% (Marvelon®) in HDL (free and total cholesterol) fraction. This can be interpreted as a favourable alteration in the LDL:HDL cholesterol ratios for the triphasic and desogestrel OC pills. Table IV summarises lipid and lipoprotein profiles for all 3 OCs at the end of one year of treatment.

Table I. Serum lipids at 0,6 and 12 months of treatment

	Microgynon®30			Triquilar®			Marvelon®		
	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth
Triglycerides									
n	34	29	25	39	29	23	32	23	20
Mean (mmol/l)	0.945	0.943	0.997	0.825	1.063	1.072	1.001	1.258	1.413
SD	0.463	0.348	0.446	0.371	0.449	0.644	0.508	0.591	0.567
% diff from -Pre mean	100	100	106	100	129	130	100	126	141
P	-	NS	NS	-	p<0.05	NS	-	NS	<0.01
Total Cholesterol									
n	34	29	25	38	29	23	32	23	20
Mean (mmol/l)	4.908	4.443	4.638	4.766	4.714	4.805	4.793	4.592	4.769
SD	0.975	1.131	0.849	0.760	0.793	0.764	0.883	0.889	0.758
% diff from -Pre mean	100	91	95	100	99	101	100	96	99
P	-	NS	NS	-	NS	NS	-	NS	NS
Free Cholesterol									
n	33	27	25	38	28	22	31	22	20
Mean (mmol/l)	1.326	1.293	1.251	1.275	1.243	1.296	1.316	1.251	1.312
SD	0.308	0.250	0.248	0.220	0.229	0.235	0.220	0.207	0.245
% diff from -Pre mean	100	98	94	100	98	102	100	95	100
P	-	NS	NS	-	NS	NS	-	NS	NS

Microgynon®30 = 30µg ethinyl oestradiol (EE) + 150µg levonorgestrel (LNG)
Marvelon® = 30µg EE + 150µg desogestrel
Triquilar® = 30µg EE + 50µg LNG x 6 days; 40µg EE + 75µg LNG x 5 days; 30µg EE + 125µg LNG x 10 days

Table II. Serum triglyceride lipoprotein fractions at 0,6, and 12 months of treatment

	Microgynon®30			Triquilar®			Marvelon®		
	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth
VLDL									
n	34	29	25	35	29	23	32	23	20
Mean (mmol/l)	0.534	0.472	0.501	0.450	0.543	0.535	0.593	0.680	0.770
SD	0.336	0.230	0.269	0.282	0.303	0.460	0.426	0.423	0.404
% diff from -Pre mean	100	88	94	100	121	119	100	115	130
P	-	NS	NS	-	NS	NS	-	NS	NS
LDL									
n	34	29	25	35	29	23	32	23	20
Mean (mmol/l)	0.170	0.183	0.177	0.145	0.187	0.193	0.167	0.220	0.221
SD	0.066	0.053	0.061	0.049	0.050	0.072	0.084	0.092	0.058
% diff from -Pre mean	100	108	104	100	130	133	100	132	132
P	-	NS	NS	-	<0.01	<0.01	-	<0.05	< 0.01
HDL									
n	34	29	25	35	29	23	32	23	20
Mean (mmol/l)	0.145	0.151	0.158	0.127	0.167	0.172	0.152	0.212	0.240
SD	0.047	0.041	0.068	0.030	0.039	0.055	0.046	0.067	0.062
% diff from -Pre mean	100	104	109	100	131	135	100	140	158
P	-	NS	NS	-	<0.001	<0.001	-	<0.001	<0.001

VLDL = very low density lipoproteins
LDL = low density lipoproteins
HDL = high density lipoproteins

Table III. Serum total and free cholesterol lipoprotein fractions at 0,6, and 12 months of treatment

	Microgynon®30			Triquilar®			Marvelon®		
	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth
VLDL									
Total cholesterol VLDL									
n	34	29	25	35	29	23	32	23	20
Mean (mmol/l)	0.754	0.617	0.608	0.609	0.631	0.573	0.843	0.818	0.810
SD	0.395	0.289	0.271	0.251	0.249	0.272	0.377	0.410	0.287
% diff from									
-Pre mean	100	82	81	100	104	94	100	97	96
P	-	NS	NS	-	NS	NS	-	NS	NS
Free cholesterol VLDL									
n	33	27	25	34	28	22	31	22	20
Mean (mmol/l)	0.253	0.235	0.228	0.225	0.231	0.213	0.298	0.303	0.296
SD	0.121	0.103	0.094	0.101	0.095	0.122	0.124	0.111	0.093
% diff from									
-Pre mean	100	93	90	100	103	95	100	102	99
P	-	NS	NS	-	NS	NS	-	NS	NS
LDL									
Total cholesterol LDL									
n	34	29	25	35	29	23	32	23	20
Mean (mmol/l)	2.536	2.459	2.371	2.583	2.395	2.344	2.409	2.129	2.221
SD	0.610	0.603	0.714	0.670	0.697	0.565	0.665	0.881	0.646
% diff from									
-Pre mean	100	97	94	100	93	91	100	88	92
P	-	NS	NS	-	NS	NS	-	NS	NS
Free cholesterol LDL									
n	33	27	25	34	28	22	31	22	20
Mean (mmol/l)	0.684	0.686	0.643	0.690	0.670	0.627	0.663	0.582	0.596
SD	0.163	0.157	0.214	0.196	0.281	0.148	0.153	0.219	0.169
% diff from									
-Pre mean	100	100	94	100	97	91	100	88	90
P	-	NS	NS	-	NS	NS	-	NS	NS

(continued)

Table III. Serum total and free cholesterol lipoprotein fractions at 0,6, and 12 months of treatment

(continued)

	Microgynon®30			Triquilar®			Marvelon®		
	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth
HDL									
Total cholesterol HDL									
n	34	29	25	35	29	23	32	23	20
Mean (mmol/l)	1.360	1.242	1.283	1.401	1.401	1.520	1.347	1.440	1.691
SD	0.306	0.198	0.239	0.257	0.218	0.211	0.320	0.269	0.767
% diff from									
-Pre mean	100	91	94	100	100	109	100	107	126
P	-	NS	NS	-	NS	NS	-	NS	NS
Free cholesterol HDL									
n	33	27	25	34	28	22	31	22	20
Mean (mmol/l	0.315	0.301	0.303	0.326	0.323	0.367	0.315	0.331	0.380
SD	0.102	0.104	0.061	0.064	0.065	0.077	0.094	0.084	0.079
% diff from									
-Pre mean	100	96	96	100	100	113	100	105	21
P	-	NS	NS	-	NS	<0.05	-	NS	<0.05

Table IV. Summary table of lipid and lipoprotein profiles at the end of 12 months of treatment

	Microgynon®30	Triquilar®	Marvelon®
Serum lipids			
Triglycerides	no change	↑	↑**
Total cholesterol	no change	no change	no change
Free cholesterol	↓	no change	no change
Triglyceride lipoproteins			
TG VLDL	no change	↑	↑
TG LDL	no change	↑**	↑**
TG HDL	↑	↑***	↑***
Total cholesterol lipoproteins			
T Chol VLDL	↓	no change	no change
T Chol LDL	↓	↓	↓
T Chol HDL	↓	↑	↑
Free cholesterol lipoproteins			
F Chol VLDL	↓	no change	no change
F Chol LDL	↓	↓	↓
F Chol HDL	no change	↑*	↑*

* = $P < 0.05$
** = $p < 0.01$
*** = $p < 0.001$

Table V shows the changes in mean glucose values after an oral glucose load with treatment with each of the three OC pills. As expected, there is a deterioration in glucose tolerance with all three OC pills.

At 12 months of treatment, the impairment caused by Marvelon® and Triquilar® (Fig.1) appears greater than that produced by Microgynon®30.

Discussion

Combined oral contraceptive pills, through their metabolic actions, increase the risk of cardiovascular disease. The use of high dose progestogen dominant OCs induces a potentially atherogenic lipoprotein profile (high LDL-cholesterol: HDL-cholesterol ratio) mostly attributed to the antioestrogenic action of the progestogen in these OCs. Caucasian studies (14) have shown that low-dose pills, both monophasic and multiphasic, produce fewer adverse effects on lipoproteins. Pharmaceutical firms introducing new progestogen combined OCs have also claimed their respective 'pill' to be superior in these respects. These claims, and the paucity of figures in Asian women (whose diet and lifestyle are different from that of their Caucasian counterparts), led to this comparative study.

There are many metabolic factors (as opposed to direct risk factors such as age, smoking, obesity, etc) affecting one's propensity to developing heart disease. Some of these are plasma cholesterol, triglycerides, their lipoprotein carriers (eg., chylomicrons, VLDL, LDL, HDL) and apolipoproteins A_1, A_2 and B. In general, high levels of LDL-cholesterol and low levels of HDL-cholesterol are potentially atherogenic and disease causing. Even among the HDL-cholesterol fraction, decreased levels of subfraction HDL_2-cholesterol correlates better with frequency and severity of ischaemic heart disease than does HDL-cholesterol or HDL_3-cholesterol subfraction.

The association between hypertriglyceridemia (mainly VLDL-triglycerides) and ischaemic heart disease, although well recognised, is not a totally direct one; it appears to be associated with the precursor-product relationship between VLDL and LDL and the inverse correlation between VLDL and LDL. A good review of these complex relationships can be found in Gaspard's paper (14). The present study measures only the major lipids and lipoprotein fractions and thus provides only a crude measure of the metabolic risks of the 3 OCs studied.

With regard to serum lipids, Triquilar® and Marvelon® produced signficant rise in triglycerides while Microgynon®30 had no appreciable change. Serum cholesterol was not significantly different for the 3 OCs. Microgynon®30 also did not appreciably alter triglyceride lipoprotein fractions over one year of treatment and caused a slight fall in both LDL- and HDL-cholesterol levels. It also caused the least derangement in glucose tolerance. These changes make it the most neutral of the 3 OCs studied in our local Chinese women.

Table V. Mean glucose (fasting, 30,60,90 and 120 minutes) at 0,6 and 12 months with the 3 OC pills after 50gm oral glucose load

	Microgynon®30			Triquilar®			Marvelon®		
	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth
Glucose Fasting									
n	34	29	25	39	29	23	32	23	19
Mean (mmol/l)	5.047	4.816	4.749	4.991	4.739	4.727	5.047	4.950	4.853
SD	0.594	0.390	0.404	0.496	0.283	0.444	0.372	0.566	0.480
% diff from -Pre mean	100	95	94	100	95	95	-	98	96
P	-	NS	<0.05	-	<0.05	<0.05	-	NS	NS
Glucose 30 min									
n	34	29	25	39	29	23	32	22	18
Mean (mmol/l)	7.218	7.362	7.651	7.544	7.596	7.566	7.492	7.881	8.019
SD	1.145	0.885	1.039	1.227	1.151	1.654	1.177	1.493	1.354
% diff from -Pre mean	100	102	106	100	101	100	100	105	107
P	-	NS	NS	-	NS	NS	-	NS	NS
Glucose 60 min									
n	34	29	25	39	29	23	32	23	19
Mean (mmol/l)	7.207	7.616	7.950	7.563	8.348	7.896	7.294	8.088	8.590
SD	1.921	1.271	1.515	1.781	1.411	2.206	1.401	1.771	1.646
% diff from -Pre mean	100	106	110	100	110	104	100	111	118
P	-	NS	NS	-	<0.05	NS	-	NS	<0.01

(continued)

Table V. Mean glucose (fasting, 30,60,90 and 120 minutes) at 0,6 and 12 months with the 3 OC pills after 50gm oral glucose load

(continued)

	Microgynon®30			Triquilar®			Marvelon®		
	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth	0 mth	6 mth	12 mth
Glucose 90 min									
n	34	29	25	39	29	23	32	22	19
Mean (mmol/l)	6.858	7.276	7.155	6.748	7.672	7.705	6.602	7.909	8.030
SD	1.644	1.270	1.455	1.556	1.598	2.294	1.306	1.660	1.756
% diff from -Pre mean	100	106	104	100	114	114	100	120	122
P	-	NS	NS	-	<0.05	NS	-	<0.01	<0.01
Glucose 120 min									
n	34	29	25	39	29	23	32	23	19
Mean (mmol/l)	6.301	6.768	6.408	6.123	6.959	7.115	6.306	6.761	7.485
SD	1.233	1.502	1.214	1.521	1.283	1.604	1.525	1.920	0.933
% diff from -Pre mean	100	107	102	100	114	116	100	107	119
P	-	NS	NS	-	<0.05	<0.05	-	NS	<0.01

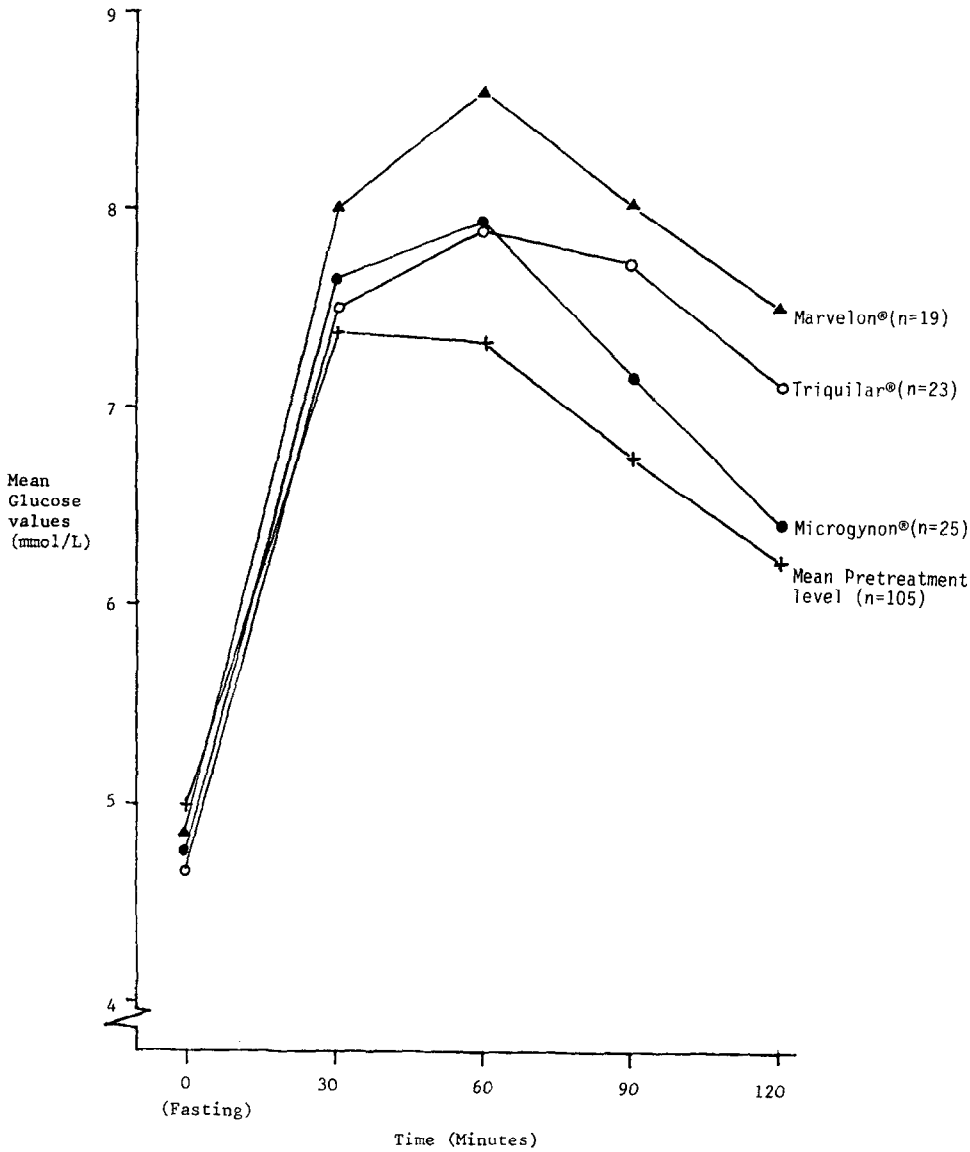

Fig.1. Glucose tolerance (as reflected by mean glucose levels) at 12 months of treatment with 3 OC pills compared with pretreatment mean glucose levels.

Marvelon® and Triquilar® cause rises in HDL-triglyceride and HDL-cholesterol and falls in LDL-cholesterol levels, changes which are considered favourable. Marvelon® however causes a larger deterioration in glucose tolerance whereas Triquilar® causes marginal deterioration in glucose tolerance when compared to that of Microgynon®. These findings are consistent with other studies on gonane progestagens(14).

The clinical significance of the metabolic changes remains to be seen in our local population. At present there is insufficient reason to switch to the newer generation OCs in the woman who is currently on the 'older' low-dose combined OC pill without problems. The assumption that levonorgestrel derivatives are "too potent" is refuted by the observation of favourable metabolic results obtained with OCs containing very low dose of progestogens. It must also be stressed that beyond their effects on carbohydrate and lipid metabolism, OCs display many other actions and it is therefore essential to have complete and in-depth studies of any newly introduced oestrogen/progestogen combination in different populations before any confident pronouncement on its safety can be made.

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